

- Q.25 Differentiate between conventional machines and CNC machines. (CO2)
- Q.26 Explain part programming formats. (CO2)
- Q.27 Explain basic components of NC machines. (CO2)
- Q.28 Enlist the joints of a robot. (CO3)
- Q.29 Describe advantage and disadvantage of LBM. (CO2)
- Q.30 Explain any one indexing methods. (CO5)
- Q.31 Prepare simple Part program for Turning. (CO5)
- Q.32 State the working principle of ECM. (CO1)
- Q.33 Explain Part program structure with example. (CO4)
- Q.34 Differentiate between NC machines and conventional machines. (CO2)
- Q.35 Explain the functions of Transducer in USM. (CO2)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Explain the principle and process of EBM and state its limitations. (CO3)
- Q.37 Explain the following milling cutters with neat sketch. (CO5)
- i) Plain milling cutter ii) End milling cutter
- Q.38 Explain in details the various components of robot. Also give the main advantages and disadvantages of robots in industries. (CO4)

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5th Sem / Auto Subject:-Advanced Manufacturing Processes

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Numerical control (CO1)
- applies only to milling machines
 - is a method for producing exact number of parts per hour
 - is a method for controlling by means of set of instructions
 - none of the mentioned
- Q.2 Ultrasonic machining is based upon (CO3)
- vibrational waves of zero frequency
 - Vibrational waves of low frequency
 - Vibrational waves of high & low frequency
 - vibrational waves of high frequency
- Q.3 Computerized numerical control machines use (CO1)
- Digital data & a computer
 - A computer and a CAM program
 - A computer, a CAM program & a digital data
 - None

- Q.4 Computer will perform the data processing functions in (CO1)
 a) VMC b) CNC
 c) DN d) all of the mentioned
- Q.5 A CNC Machines uses (CO1)
 a) Open loop system b) Closed loop system
 c) Both (a) & (b) d) None
- Q.6 G code for Linear interpolation (CO3)
 a) G01 b) G02
 c) G03 d) G70
- Q.7 Which of the following is not the advantage of CNC machines? (CO3)
 a) Reduced scrap rate
 b) Improved strength of the components
 c) Higher flexibility
 d) Improved quality
- Q.8 Which of the following system has feedback system? (CO2)
 a) Open loop system b) Closed loop system
 c) Direct loop system d) None of the above
- Q.9 Laser is produced by (CO1)
 a) Diamond b) Ruby
 c) Graphite d) Magnesium
- Q.10 NC contouring is an example of (CO2)
 a) Continuous path positioning
 b) Point-Point positioning
 c) Absolute positioning
 d) Incremental positioning

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Name the M code for “spindle stop” and “tool Change” (CO3)
- Q.12 LASER Stands for _____ (CO3)
- Q.13 Write full form of ECM. (CO3)
- Q.14 Conversion of units is possible in case of CNC machines (True/False) (CO3)
- Q.15 DNC stands for _____ (CO3)
- Q.16 Define Point to point Motion control.
- Q.17 Write the use of collets in milling Machine.
- Q.18 Write full form of MCU.
- Q.19 First milling machine came into existence in about 1770 (True/False) (CO4)
- Q.20 Cutting a slot in the head of a screw or bolt is the example of milling machine. (True/False) (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain Linear and circular interpolation.
- Q.22 Explain the following milling operations. (Co5)
 i) Straddle milling ii) End milling
- Q.23 Describe M00, M01, M10, M17, code in part programming. (CO3)
- Q.24 State the Applications of Electro chemical Machining. (CO1)

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